7ai	Power dissipated in the resistor is due to the non-zero root-mean-square current. Accept: I^2 is always positive, therefore P(=I^2R) is always positive.
7aii	either maximum power = $I_0^2 R$ or average power = $I_{RMS}^2 R$ $I_0 = \sqrt{2} \times I_{RMS}$ maximum power = 2 × average power Thus, ratio = 0.5

7bi	Improve flux linkage between primary and secondary coil, and strengthen the magnetic flux density.
7bii	Magnetic flux in the core is changing, causing induced e.m.f. or current in the core. Current in core causes heating effect.
7biii	The iron core is made of laminated sheets to reduce power loss due to induced current.